

Firm-level exposure to trade policy shocks: A multidimensional measurement approach

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Abstract

We propose a firm-level measure of exposure to trade policy shifts that combines characteristics (tradability of goods, share of output exported and corporate risk disclosures) with information from stock returns. We show that the measure reliably captures out-of-sample differences in price responses and sentiment related to trade tensions, both in US and international data. Differences across firms are economically important with return effects of 140 bp around tariff announcements. We argue that such a multidimensional measure is a useful tool for future research on trade policy risk.

KEYWORDS

equities, exposure measurement, international trade, risk exposures, trade policy

JEL CLASSIFICATION

F13, G10

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1 | INTRODUCTION

Trade policy influences the financial performance of firms. It affects the cost and availability of foreign inputs, access to export markets and the extent of foreign competition. Shifts in trade policy will have heterogeneous effects across firms (e.g., Fajgelbaum et al., 2020). For example, more productive firms tend to thrive with trade liberalisation while less productive firms benefit from protectionism (Bernard et al., 2006; Melitz, 2003).

This paper analyses a firm-level measure of exposure to shifts in trade policy and documents it captures substantial heterogeneity across firms in their reaction to trade policy shifts. This measure combines multiple dimensions of trade policy exposure and exploits information in stock return covariance in addition to industry and firm characteristics. We show in detailed validation tests that the measure captures heterogeneous exposure of firms to trade policy shocks in terms of both stock price reactions and sentiment or uncertainty on trade policy. Importantly, the measure is available ex-ante at the firm level and effectively captures heterogeneous responses ex-post. We also show that the measure is not subsumed by information contained in standard firm classifications into size categories or industries.

Such a measure can be used in several academic and practical applications. For example, the impact of changes in trade policy needs to be well understood by policy makers. The exposure measure allows to identify which areas of the economy will benefit or suffer from proposed changes. Furthermore, accurate measurement of exposure to shifts in trade policy is needed in asset pricing research to test whether this is a priced risk. It also allows to construct investment portfolios that enable investors to manage trade policy risk, be it to harvest a potential premium or to hedge undesired exposures in nontradable components of their wealth.

Previous research has used several measures to capture firms' exposures to shifts in trade policy. Barrot et al. (2019) use shipping costs for a firm's goods, following Bernard et al. (2006), to measure exposure to globalisation. High shipping costs are structural economic barriers that decrease the potential for international trade. This low tradability will make it less likely for a firm to sell its goods internationally. Therefore, firms producing products with high shipping costs, or low tradability, are less exposed to international trade flows and the risk of shifts in trade policy. Furthermore, Tian (2018) distinguishes between firms based on the proportion of their output that is exported. Firms which export a larger fraction of their output are more exposed to changes in international trade policy. Yet another approach to capture exposure to shifts in trade policy is to use textual analysis of a firm's risk disclosures (see Baker et al., 2016; Caldara et al., 2020). A firm's management could indicate themselves that the firm is exposed to the risk of the introduction or removal of trade barriers.

We propose a multidimensional measure of exposure to shifts in trade policy by combining these three individual measures.¹ Furthermore, we also include information about stock return covariances. Stock returns allow to capture relevant firm-level information based on the assessment of market participants. Stocks with a high exposure according to this measure are expected to perform well when trade barriers decrease, so we call them trade peace stocks.

¹We note that it could be desirable for some applications to omit certain components of our measure. For example, textual analysis that identifies stocks exposed to trade policy risk will lead to false positives that can be easily identified by human analysts. Even if such a measure adds information on average, occurrence of false positives might not be acceptable for investors who conduct an analysis of their portfolio holdings. This could motivate removing the textual analysis component in investment applications.

Similarly, stocks with a low exposure are expected to perform relatively well when trade barriers increase and are called trade war stocks.

Compared with the measures proposed previously, our multidimensional measure has the advantage that it uses a broader set of information. International trade impacts firms via different, interdependent channels (Bernard et al., 2018). Given this complexity, we cannot expect that a single source of information captures exposure to shifts in trade policy accurately. Furthermore, combining multiple measures reduces the impact of noise present in each of them individually (see Kahneman et al., 2021). Therefore, we expect that a multidimensional measure captures exposure to trade policy risk more accurately. This paper provides empirical evidence supporting this idea. A related advantage of combining several measures is that results of empirical analysis will not be overly dependent on variable selection choices.

To ensure that proxies of trade policy exposure effectively capture the economic risks they target, we conduct tests that assess heterogeneous effects of trade policy shocks across firms. When using a reliable measure of exposure to group firms by their expected sensitivity to trade policy shocks, high exposure firms should be significantly more negatively impacted than low exposure firms by out of sample shocks. Such cross-sectional validation of exposure measures is an important step before analysing further economic differences across high and low exposure firms. We follow a double validation procedure, assessing heterogeneity of firm-level effects of trade policy events in terms of both stock price reactions and discussions on trade policy between management and analysts.

We analyse the effect of trade policy events on stock prices via the abnormal return around tariff announcements between the United States and China, following Amiti et al. (2020). Abnormal returns can be interpreted as an ex-post measure of exposure to trade policy shifts,² and are commonly used in the literature on international trade (e.g., Davies & Studnicka, 2018; Fisman et al., 2014; Greenland et al., 2020).³ We show that our multidimensional exposure measure (which is based on data known before the events) significantly explains cross sectional variation in the abnormal returns observed around these events. Moreover, each of the individual dimensions has incremental explanatory power over the other two in these tests. At the same time, none of them add value when controlling for the combined measure. This indicates that a combination of these measures is useful to capture exposure to trade policy shifts.

These tests allow us to document statistically significant heterogeneity across firms. Trade peace stocks have on average a negative cumulative abnormal return of around 70 bp from the day of the tariff announcement until 1 week after. Trade war stocks, on the other hand, have a positive cumulative abnormal return of around 70 bp in the same time window. Apart from their statistical significance, these results are also large in terms of economic magnitude. Indeed, the 140 bp difference in abnormal returns between trade peace and trade war stocks represents more than 10 times the average weekly excess return of the US equity market, which is 12 bp, and almost two thirds of the 222 bp weekly standard deviation of market returns.⁴ The average cumulative abnormal returns are also large relative to the impact on stock returns of

²Compared with using the abnormal returns directly as an exposure measure, as in Greenland et al. (2020), our measure is available ex-ante. Furthermore, abnormal returns are a noisy and purely empirical measure, whereas our exposure measure is based on fundamental economic mechanisms.

³Abnormal returns are also used to study the impact on the equity market of other events, such as disasters. See for instance Ding et al. (2021) and Pagano et al. (2022), who study the impact of the recent pandemic outbreak.

⁴We estimate the average and standard deviation of weekly excess equity market returns in the United States using the data from the Kenneth French Data library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html) over the period from the 1st of July 1963 to the 30th of June 2020.

other important events, such as the pre-announcement effect of Federal Open Market Committee (FOMC) meetings.⁵

To assess heterogeneity of firm-level sentiment and uncertainty on trade policy, we analyse text in quarterly earnings calls.⁶ We identify text that discusses trade policy topics and measure the surrounding sentiment and uncertainty, following the methodology of Hassan et al. (2019) and Caldara et al. (2020). Earnings calls include a presentation by the firm's management and a Q&A session with external analysts. These measures therefore capture both internal and external perspectives on the impact of trade policy on the firm. In cross-sectional regressions, we find that trade peace firms tend to have a significantly more negative trade sentiment and higher trade uncertainty than trade war firms in times of rising trade tensions. In the cross-section, a one standard deviation increase in exposure corresponds to a decrease in sentiment of one-tenth of a standard deviation, and an increase in uncertainty of around one-fifth of a standard deviation. To put these numbers into perspective, the decrease in sentiment is more than half the drop on average trade sentiment observed during the 2018 spike in trade tensions.

Overall, this double validation procedure provides support that the proposed measure captures trade policy exposure. The results also document substantial heterogeneity of effects across firms. These results hold both in US and international data. In addition, we show that standard firm classifications by size and industry are insufficient to capture such heterogeneity.

This paper is related to the literature on international trade, and more specifically to research that considers heterogeneity among firms. Melitz (2003) and Bernard et al. (2003) propose models that use differences in productivity to explain why some firms export while others don't. They show that a reduction in trade barriers benefits larger, more efficient firms, who tend to be exporters, while hurting the smaller, least efficient ones, who only serve the domestic market. Chaney (2008) expands these models by considering multiple countries with asymmetric trade barriers. Eaton et al. (2004, 2011) also consider multiple export markets. Again, they show that lower trade barriers benefit large exporters while the corresponding import competition results in the exit of small, domestic firms. Instead of focusing on exports, Antràs et al. (2017) analyse firms' sourcing decisions. As was the case for exports, there exists much heterogeneity across firms. Larger, more efficient firms are more inclined to import inputs. They point out that import decisions are more complex and interdependent than export decisions, due to their impact on marginal costs. Bernard et al. (2018) propose a framework in which export and sourcing decisions are combined. They also include other considerations, such as the choice between exporting and producing in foreign countries via foreign direct investment (Helpman et al., 2004), to model the environment of large global firms in more detail. This literature highlights the heterogeneity in the impact that firms experience from changes in trade barriers. We contribute by focussing on a practical measure to capture this heterogeneity and show that it subsumes the information contained in firm size, which is identified by these theories as being related to international trade exposure.

To construct our measure of trade policy risk exposure, we draw on a literature that relates international trade exposure to properties of stock returns. For example, Tian (2018) finds that firms with a high proportion of output that is exported have more cyclical asset returns. Hoberg and Moon (2019) find that firms selling output abroad tend to have higher stock returns while firms purchasing input abroad have lower stock returns. This is consistent with the idea that offshore output is a source

⁵For instance, Lucca and Moench (2014) find that in the hours preceding the FOMC announcements, on average the S&P 500 experiences an abnormal return of 0.5%.

⁶We discuss in the validation section that analyst earnings calls contain useful data to measure sentiment ex-post but not to measure exposure ex-ante.

of risk, while offshore input can be used as a hedge. Barrot et al. (2019) measure exposure to international trade via shipping costs and find that exposed firms earn a risk premium. They point to the risk of increased import competition to explain the results. Huang et al. (2023) study stock price reactions to the US-China trade war and construct exposure measures based on detailed firm-level data on imports from and exports to China, including product-level data and supply chain linkages. Their finding that a one standard deviation increase in revenue from China corresponds to a 48 bp lower average return around a tariff announcement is in line with the 60 bp reduction we capture for a standard deviation increase in our exposure measure, which is not tailored to this specific event. We also draw on methods of text analysis to capture exposure to trade policy risk. We exploit information in 10-K filings to construct our ex-ante exposure measure, and information extracted from earnings call transcripts for ex-post validation (e.g., Hassan et al., 2019).⁷ We argue that our exposure measure is more suitable for ex-ante exposure measurement compared with a measure based on earnings call transcripts, as proposed by Hassan et al. (2019), due to strong time variation in attention to trade policy in discussions between analysts and management. We build on this literature and combine information from existing measures of trade policy exposure, as well as from stock return covariance.

A related strand of literature shows that foreign exposure of firms is an important driver of heterogeneous reactions of firms to changes in government policies more broadly. Wagner et al. (2018a) study stock price reactions to the 2016 US presidential election and find that more internationally oriented firms, for example, measured by the percentage of foreign revenues or profits, experience a more negative stock price reaction to the election. They remain agnostic about whether this effect is mainly driven by changing expectations on trade or tax policy. Wagner et al. (2018b, 2020) and Gaertner et al. (2020) focus on the heterogeneous impact of the Tax Cuts and Jobs Act introduced in 2017 across firms and also document a relation with the degree of international exposure of firms. The exposure measures in this line of work tend to capture the importance of foreign operations of a firm, but not necessarily the extent of international trade flows. Our exposure measure, which specifically captures exposure to international trade as opposed to foreign operations more broadly, could contribute to this literature and indicate whether stock price reactions are mostly due to changing expectations regarding trade or other policies.

This paper provides three contributions to the literature. First, we propose a multidimensional measure of exposure to shifts in trade policy. The measure combines previously proposed metrics, while adding information from stock return covariances. We argue that, due to the complex nature of international trade, including multiple dimensions helps to accurately capture exposures. Importantly, the measure is ex-ante available at the firm level, which makes it suitable for a wide range of applications. Second, we conduct validation tests that focus on the cross-sectional variation of exposure to shifts in trade policy. Our double validation procedure makes use of ex-post reactions to trade policy shocks both in stock prices and in discussions on trade between management and analysts. The use of sentiment and uncertainty from text data is a valuable addition compared with previous work and allows testing whether the results based on stock returns are supported by an independent data source. Third, armed with a comprehensive measure of risk exposure and a framework for assessing reactions to trade policy events, we document significant heterogeneity in firms' reactions to tariff announcements related to the US-China trade war and global trade events. Again, we document this heterogeneity both in terms of stock returns and discussions between management and analysts.

⁷Other relevant work in this area includes Baker et al. (2016), who construct a measure of economic policy uncertainty, and Caldara et al. (2020), who focus on trade policy uncertainty specifically.

The remainder of this paper proceeds as follows. Section 2 describes how the trade policy exposure measure is constructed. Section 3 assesses firm-level heterogeneity in terms of stock price reactions and trade sentiment and uncertainty extracted from earnings calls. Section 4 contains the results based on international data. Finally, Section 5 concludes.

2 | MEASURING EXPOSURE TO SHIFTS IN TRADE POLICY

This section outlines the construction details of our firm-level measure of exposure to shifts in trade policy. We draw on firm characteristics proposed in the literature: tradability, export share and trade risk disclosures. These characteristics should capture the economic drivers of firms' sensitivity to shifts in trade policy. To alleviate issues in granularity and coverage that each of these metrics face, we augment firm characteristics with covariances of stock returns with the respective mimicking portfolio. This approach allows using information contained in stock returns to enrich the information that would be available from characteristics alone. To account for the multidimensional nature of trade policy exposure, we then combine those three dimensions. Our main analysis is based on a US stock universe consisting of the 1000 largest firms traded on the New York Stock Exchange or Nasdaq. The universe is reconstituted quarterly on the third Friday of March, June, September and December.⁸ When referring to quarters in this paper, these are understood to start and end on these days. This universe of large and liquid stocks ensures results are relevant for investment applications.

2.1 | Tradability

We obtain the annual data to construct our estimate of shipping costs from the US Census Bureau. Similar to Barrot et al. (2019), we define industry-level shipping costs based on the cost, insurance and freight (CIF) import value and customs import value data fields, as follows.

$$\text{Shipping costs} = \frac{\text{CIF Import Value}}{\text{Customs Import value}} - 1.$$

Shipping costs therefore captures the cost of insurance and freight to transport goods internationally, as a percentage of the value of the goods.

This data is classified into industries based on the North American Industry Classification System (NAICS). If available, we assign a value to industries at the most granular six-digit level industries. If the required data is not available at this level, we use the value for the corresponding four-digit level industry group, or the three-digit level subsector for the industry. If data are unavailable at this level, we consider the value missing. Every June, a stock is assigned the shipping cost value of its corresponding industry⁹ based on data of the previous year. Note that shipping costs capture the

⁸More specifically, we use data for the Scientific Beta extended US universe. The detailed construction process for this universe can be found in documentation accessible via www.scientificbeta.com.

⁹Note that the universe data for the Scientific Beta extended US universe available to us contains NACE industry classification codes. Therefore, we first map the NAICS 6-digit level industries to NACE codes, via publicly available industry mappings from ISIC to NAICS and ISIC to NACE.

opposite notion of tradability, high shipping costs means that a firm has low tradability. Consequently, when considering tradability, we rank stocks according to the negative of the shipping costs value.

A firm's tradability is based on the goods it produces. Therefore, the two key channels via which it captures exposure to shifts in trade policy are the possibility to sell the good abroad and competition from foreign firms selling similar goods. It does not necessarily capture the extent to which inputs can be sourced from abroad. Since we focus on a universe of relatively large stocks, we expect that the former effect dominates (e.g., Bernard et al., 2003). Larger firms benefit from the export possibilities while smaller firms run more risk of being displaced by import competition. Therefore, an increase in trade barriers will tend to have a negative impact on firms due to reduced export possibilities in our universe. Tradability captures the potential for international trade in a certain industry. Firms producing goods which are easily tradable will operate in a more global environment, since shipping costs present a form of trade barriers. These firms are therefore more exposed to the effects of shifts in trade policy. Consequently, if the government enacts a policy that results in increased trade barriers, tradable firms will suffer relatively more than nontradable firms because they relied more on overseas export. On the other hand, a reduction in trade barriers will benefit tradable firms relatively more because their potential for international trade is larger.

Clearly, shipping costs only capture one aspect of exposure to shifts in trade policy. While it captures a structural economic feature of an industry that gives rise to a natural trade barrier, other hurdles such as time to ship, information barriers, or contract enforcement costs are not captured (Barrot et al., 2019). Another important drawback is that shipping costs are only available for goods. Therefore, the measure does not apply to firms in service sectors.

2.2 | Export share

For the second characteristic we follow Tian (2018) and use data from input-output tables. Specifically, we download annual input-output tables from the OECD website for all available countries and sectors. From these tables, we obtain the value of exports and the value of the total output per sector-country pair. Using this data, we compute the export share of a sector s as follows.¹⁰

$$Export\ share_s = \frac{\sum_c Exports_{s,c}}{\sum_c Output_{s,c}}.$$

In the equation above, s,c indicates a sector-country pair and the sums are taken across all countries. The export share captures the realised amount of exports in a given sector, as a percentage of the total output in that sector. As for shipping costs, every June, a stock is

¹⁰De Gregorio et al. (1994) and Lombardo and Ravenna (2014) construct a similar measure, but also include the value of imports in the numerator. We follow Tian (2018) and include only export. Since the OECD data includes many countries, it should be representative for the global volume of international trade. Globally, total imports and exports for a sector correspond. Therefore, including imports next to exports should not add much information in terms of the ranking of each sector according to this measure. We have confirmed this is the case in our data (see results in the Supporting Information S1: Appendix).

assigned the export share value of the sector to which it belongs¹¹ based on data of the previous year.¹²

The export share captures the *realised* level of international trade, whereas tradability captures the *potential* for international trade. An increase in trade barriers will impose extra costs on firms in sectors which rely on exports for a big share of their output, while having less of an impact on nonexporting sectors. Therefore, firms in sectors with a high export share are expected to suffer more from increased trade barriers. Conversely, trade liberalisation will reduce costs for firms in sectors with a high export share while not having much impact on domestic sectors.

A benefit of this measure is that it is available for an exhaustive set of sectors that covers the full universe. However, this coverage comes with the cost that the sector grouping is somewhat coarse. The OECD sector classification distinguishes between only 36 sectors.¹³ Given that the literature has documented substantial heterogeneity in firm outcomes within industries as a result of trade liberalisation (e.g., Bernard et al., 2006), it is clear that using only the export share characteristic as a measure of exposure to shifts in trade policy would not be optimal. The information contained in the export share should therefore be complemented with other measures.

2.3 | Risk disclosures

The third characteristic is based on 10-K risk disclosure data. Following Hassan et al. (2019) and Caldara et al. (2020), we use textual analysis to extract exposures from text data. We first obtain annual 10-K filings for the firms in our universe by web scraping the SEC's EDGAR system and extract the text in Section 1A 'risk factors' from each filing. Using this risk disclosure text, we construct a measure of exposure to shifts in trade policy via a dictionary-based approach. Our trade dictionary is based on the trade-specific keywords proposed by Baker et al. (2016) and Caldara et al. (2020). It contains 23 keywords, such as *tariff*, *import duty* and *trade policy*. The full list is shown in the Supporting Information S1: Appendixx. The measure is defined as the fraction of words in the risk disclosures that are in the trade dictionary.

$$\text{Risk disclosures} = \frac{\sum_w^{W_T} (1_{w \in \text{keywords}})}{W_T}.$$

In which, W_T is the total number of words in a firm's risk disclosures and $1_{w \in \text{keywords}}$ takes the value of 1 when word w is in the trade dictionary and 0 otherwise. Manual validation checks of the text captured by the dictionary pointed out that the keyword *tariff* is often used in a context not related to international trade by telecom and utility firms. Consequently, *tariff* is omitted from the dictionary when calculating the risk disclosures score for firms in these sectors. Details on this procedure, including data collection, text preprocessing, validation

¹¹As for shipping costs, we first map the OECD sectors to NACE codes to link them to our universe. OECD sectors aggregate industries using the ISIC classification, so we can use the publicly available industry mappings from ISIC to NACE.

¹²Because we have OECD input-output tables from 2005 to 2015, following June 2016 we assign to stocks the export share value of the industry based on the most recent available year (which is 2015).

¹³This compares with 81 sectors for which we have distinct values of shipping costs, meaning that the level of granularity of export share is less than half than the one that we have for tradability.

checks and tests with alternative specifications can be found in the Supporting Information S1: [Appendix](#). Central Index Keys (CIK), the firm identifiers from EDGAR, are linked to our stock universe via tickers, using the mapping provided by the SEC.¹⁴ Every June, each stock is assigned the risk disclosures exposure score based on the text from its most recent 10-K filing.

Risk disclosures are a useful complement to the other two exposure measures. Firms are legally required to disclose material risks in their annual 10-K filings. Litigation risk thus incentivises them to provide a comprehensive overview of the risks they face. Furthermore, these disclosures should be forward-looking, which is particularly useful for our purpose of creating a measure that allows to capture risk exposures out-of-sample. Campbell et al. (2014) find that 10-K risk disclosures tend to be firm-specific and reflect the type of risks that a particular firm faces.¹⁵ This alleviates concerns that these official disclosures are mostly boilerplate language. Therefore, we expect that firms devoting a lot of attention to the discussions related to trade policy in their risk disclosures will be more sensitive to corresponding changes. Based on extensive validation readings of the text fragments that we capture, described in the Supporting Information S1: [Appendix](#), the disclosures mostly indicate that an increase in trade barriers would be harmful for the firm. Consequently, a higher fraction of risk disclosures devoted to trade policy is interpreted as indicating that the firm will suffer more due to an increase in trade barriers.

A benefit of the risk disclosures measure compared with tradability and export share is that it is available at the firm-level. It also has the potential to capture any channel via which changes in trade barriers can influence a firm, as long as analysts and management pay attention to such a channel. These benefits do come with the drawback that it is potentially noisier. As opposed to the fundamental economic features of a firm or industry captured by tradability and export share, risk disclosures contain a subjective assessment of the risks faced by the firm. Transforming the text into a numerical score also adds a layer of noise. Furthermore, a substantial fraction of firms in our sample does not mention any terms from our trade dictionary. Consequently, the risk disclosures measure is not able to distinguish between these firms.

We obtain annual exposures according to each dimension (the tradability, export share and risk disclosures measure) at each June between 2006 and 2019. Our data starts in 2006 because the SEC introduced the requirement to include Section 1A 'risk factors' in 10-K reports for fiscal years ending after 1 December 2005. Table 1 presents the average coverage and cross-sectional distribution of each of the measures over time. Tradability values are only available for around 39% of the stocks in our universe since shipping costs are not available for services. The majority of the firms for which we do have a tradability value incur shipping costs for their products between 1% and 5%. As mentioned above, export shares are available for nearly all stocks in our universe. There is a wide dispersion in export shares, with a 20th percentile value of 8.1% and an 80th percentile value of 31.5%. The summary statistics for risk disclosures show that we are able to obtain the required data for around 92% of our sample. The 20th percentile value of 0% indicates that, in all years, at least 20% of the firms do not mention any of the terms in our trade dictionary. In most years, this number is even above 40%. As a further description of what each of these measures captures, we show the industries with the highest and lowest tradability and export shares in the Supporting Information S1: [Appendix](#), together with a sample of text fragments that are captured by the risk disclosures measure.

¹⁴See <https://www.sec.gov/include/ticker.txt>.

¹⁵Beatty et al. (2019) give a note of caution here, arguing that the risk disclosures became less reflective of the underlying economic risks over time after the financial crisis of 2008.

TABLE 1 Summary statistics.

This table reports the coverage and cross-sectional distribution of the tradability, export share and risk disclosures measures. The cross-sectional distribution ignores any missing values. These values are calculated every June between 2006 and 2019 and the table shows the time-series averages. The universe consists of the 1000 largest US firms traded on the New York Stock Exchange or Nasdaq.

	Coverage	Minimum	20th percentile	40th percentile	60th percentile	80th percentile	Maximum
Tradability	39%	1.1%	1.7%	2.5%	3.3%	4.8%	20.3%
Export share	98%	0.2%	8.1%	9.4%	14.3%	31.5%	45.1%
Risk disclosures	92%	0.00%	0.00%	0.00%	0.01%	0.04%	0.49%

2.4 | Information from stock return covariances

Apart from the tradability, export share and risk disclosures measures described above, which we will refer to as characteristics from now on, we also add returns-based information to our multidimensional exposure measure. We do this by constructing mimicking portfolios based on each characteristic. These portfolios go long stocks with a high characteristic score and short stocks with a low characteristic score. Since the characteristics carry information on trade policy risk, these portfolios reflect the corresponding risk exposures. A measure of the covariance between each stock's returns and these mimicking portfolios is therefore a market-based measure of exposure to the trade policy risk dimensions represented by the characteristics. The covariance measures capture whether a stock behaves similar as stocks with a high characteristic score and therefore tease out information about exposures that is firm specific and available for all firms, even those without observed characteristics. This additional information is an important complement to the characteristics proposed by earlier research.

The goal of the mimicking portfolios is to create a time series that captures the difference in return between stocks with a high characteristic score and stocks with a low characteristic score. For tradability, we construct an equal-weighted portfolio of all stocks with shipping costs that are higher than or equal to the 80th cross-sectional percentile value and an equal-weighted portfolio of the stocks with shipping costs lower than or equal to the 20th percentile. The tradability mimicking portfolio goes long the low shipping costs portfolio and short the high shipping costs portfolio. We create similar equal-weighted portfolios based on the 80th and 20th percentile values of export share. The export share mimicking portfolio goes long the high export share portfolio and short the low export share portfolio. For risk disclosures, the large fraction of stocks with a zero score means taking the stocks below the 20th percentile isn't suitable. Therefore, the mimicking portfolio simply goes long an equal-weighted portfolio of all stocks that mention at least one term in the trade dictionary and goes short an equal-weighted portfolio of all stocks that do not mention any trade term. We thus create three mimicking portfolios for which we have daily returns starting in June 2006. All portfolios are rebalanced annually in June.

Then, we estimate the covariance score for each stock in the universe with respect to each of the mimicking portfolios. These covariance scores aim to capture whether a stock's returns behave similar as a typical stock with a high characteristic score. The first step in this estimation is to regress the returns of a stock in excess of the risk-free rate on the returns of one of the mimicking portfolios, while controlling for the market return in excess of the risk-free rate.

$$R_t^i = \alpha^{i,m} + \gamma^{i,m} \times MKT_t + \beta^{i,m} \times MIM_t^m + \varepsilon_t^{i,m}.$$

R_t^i is the excess return on stock i on day t , MKT_t is the excess return on a cap-weighted portfolio of all stocks in the universe on day t , and MIM_t^m is the return on mimicking portfolios m on day t . Every June, this regression is estimated for each stock-mimicking portfolio pair, based on the most recent 2 years of daily returns.¹⁶ This means we estimate 3000 $\beta^{i,m}$ values every June, for 1000 stocks i and three mimicking portfolios m . Importantly, we estimate this regression using data in a backward-looking window. This ensures that our final exposure measures are only based on ex-ante available information.

The second step to create the covariance score is to apply Bayesian shrinkage to all $\beta^{i,m}$ values (Vasicek, 1973). In June of year y , the covariance score for each of the three dimensions m for each stock i is calculated as follows:

$$Covariance_y^{i,m} = \beta_y^{m,prior} \times \frac{\sigma_{y,i,m}^2}{\sigma_{y,i,m}^2 + \sigma_{y,m}^2} + \hat{\beta}_y^{i,m} \times \left(1 - \frac{\sigma_{y,i,m}^2}{\sigma_{y,i,m}^2 + \sigma_{y,m}^2}\right).$$

$\beta_y^{m,prior}$ is the average of the $\hat{\beta}_y^{i,m}$ values, averaged across all stocks i , $\sigma_{y,i,m}$ is the standard error of the estimated $\hat{\beta}_y^{i,m}$ and $\sigma_{y,m}$ is the standard deviation of the $\hat{\beta}_y^{i,m}$ values, across all stocks i . We obtain these covariance scores for every June from 2008 until 2019. This data starts 2 years after the characteristic scores since we require 2 years of mimicking portfolio returns for the estimation. In the Supporting Information S1: Appendix, we show that these covariance scores are robust out-of-sample.

2.5 | A combined measure of exposure

The covariance measures extract information from stock market returns. To the extent that market participants draw on a wide range of data sources for their trading decisions, they incorporate information that goes beyond what can be captured by the data used to construct the tradability, export share and risk disclosures characteristics. Therefore, the covariance measures capture information on trade policy exposure beyond the information in characteristics alone.

Covariances also offer some more practical advantages compared with the characteristics, since they are available at the firm level for all stocks in the universe (as long as they have a sufficient return history for estimation). In contrast, tradability and export share are based on data at the level of granular industries rather than individual firms. The information content of risk disclosures is limited by the large proportion of firms that do not mention trade policy terms in their disclosures. Using characteristics alone therefore does not allow to make fine grain distinctions between firms' exposure to shifts in trade policy. Furthermore, shipping cost data and to a lesser extent risk disclosures, are not available for all stocks in the universe. The covariance measures are directly available at the firm level and thus allow for granularity and high coverage. These features are desirable for applications. For example, when managing sensitivity to shifts in trade policy in an equity portfolio, the investor would need information on all firms in the market.

¹⁶We require at least 70% of valid observations over the most recent 2 years to do the estimation.

We incorporate the advantages of the market-based covariances in our exposure measure by augmenting the characteristics. For tradability, export share and risk disclosures, we first rank each stock in the universe according to the characteristic and covariance score independently. Any tied values are given the average rank of these values and missing values are assigned the median rank. Note that for the tradability characteristic, we assign ranks in the opposite direction, since low shipping costs indicate a negative sensitivity to an increase in trade barriers. We then define augmented scores for tradability, export share and risk disclosures, as the average of the respective characteristic and covariance rank for the stock.¹⁷

The augmented scores are available for all stocks in the universe, granular and well-distributed. We replace missing characteristics with the median rank in the universe, which means we avoid giving particularly high or low ranks to firms with missing information. Since the covariance scores are unique for each stock, we also avoid the lack of granularity of the export share and risk disclosure characteristics.

We expect that combining the information in the augmented tradability, export share and risk disclosures measures will improve our ability to capture a firm's exposure to shifts in trade policy. These shifts impact firms via multiple channels. High trade barriers reduce export possibilities, foreign competition (Bernard et al., 2003) and the availability of foreign inputs (Antràs et al., 2017). Tradability and export share potentially capture export effects, but not necessarily the input channel. Risk disclosures do not target one specific channel but have the potential to include information on all of them. The information content in the three measures is therefore complementary. Given the practical advantages introduced by the addition of the covariances, in terms of granularity and coverage, we can combine the three augmented scores in a straightforward manner by simply taking the equal-weighted average.

Table 2 shows how all these different measures relate to each other. First, we select the top and bottom 30% of stocks based on the augmented score for each dimension and for the combined score. For each stock selection, we calculated the average of the exposure measures across all stocks it contains at a given point in time. We computed these exposure measures using the original characteristic, the covariance score related to the characteristic and the augmented score. This is done each year between 2008 and 2019, and the table shows the time-series averages. The reported exposure measures are standardised and thus can be interpreted as the distance from the cross-sectional average expressed in terms of standard error.

We highlight three key observations from this table. First, the augmenting procedure preserves the information content in both the characteristics and the covariances. When conditioning on high and low augmented tradability scores, the average tradability characteristic is 66% above and 69% below average, and the average tradability covariance is 95% above and 91% below average, respectively. For export share and risk disclosures these values are roughly similar. This means that our methodology successfully creates a well-distributed augmented measure that combines the information in the characteristics and covariances. Second, the three dimensions indeed seem to capture different information. For example, when conditioning on low augmented tradability scores, the average export share characteristic is around the average of the full universe. Similarly, the average tradability characteristic for stocks with a high augmented risk disclosure score is only 9%

¹⁷In the end, we are interested in how a stock reacts to a change in trade policy, so using only the returns-based covariance scores may also seem appealing. However, an extra layer of estimation risk is introduced when obtaining the covariance scores. Therefore, we give an equal weight to the more fundamental information in the characteristics as an additional form of shrinkage. Nonetheless, we show in the Supporting Information S1: Appendix that our main conclusions based on the augmented scores also hold for both characteristics and covariances individually.

TABLE 2 Conditional average exposure measures.

This table reports the conditional average values of the characteristic, covariance and augmented scores for each of the tradability, export share and risk disclosures dimensions. The averages are calculated for stock selections based on the condition indicated in the column headers, that is, for stocks with either a high or a low augmented tradability, augmented export share, augmented risk disclosures, or combined exposure score. High and low selections are defined as stocks with a score higher than or equal to the 70th percentile, and lower than or equal to the 30th percentile, respectively. Every June between 2008 and 2019, each measure is first transformed to a z-score in the cross-section. Then the conditional averages are calculated. The values shown in the table are the time-series averages of these annual conditional averages. The universe consists of the 1000 largest US firms traded on the New York Stock Exchange or Nasdaq.

	Tradability		Export share		Risk Disclosures		Combined	
	High	Low	High	Low	High	Low	High	Low
Tradability								
Characteristic	66%	−69%	16%	−42%	9%	−23%	48%	−107%
Covariance	95%	−91%	−14%	−6%	7%	−19%	46%	−36%
Augmented	117%	−117%	10%	−10%	9%	−16%	74%	−44%
Export share								
Characteristic	50%	−2%	116%	−89%	61%	−62%	92%	−75%
Covariance	23%	4%	92%	−104%	57%	−82%	68%	−94%
Augmented	36%	6%	119%	−121%	71%	−90%	92%	−104%
Risk disclosures								
Characteristic	−5%	20%	37%	−38%	93%	−51%	49%	−43%
Covariance	44%	−14%	73%	−99%	89%	−108%	86%	−106%
Augmented	29%	3%	73%	−90%	127%	−115%	94%	−102%

above the universe average. The export share and risk disclosures dimensions do seem more related to each other. Still, this indicates that combining the three dimensions can add value over using only a single one. Third, the combined exposure score is well-balanced in the sense that it captures the information from each dimension without giving an excessive weight to a single one. Stocks with a high combined exposure score have an average augmented tradability, export share and risk disclosures score that is 74%, 92% and 94% above the universe average, respectively. Similarly, stocks with a low combined exposure score have augmented scores that are on average 44%, 104% and 102% below the universe average. The absolute values for the characteristics and covariances are a little lower in most cases, but still indicate that the combined score captures the information contained in each measure in a balanced way.

3 | VALIDATION OF THE EXPOSURE MEASURES

So far, we have proposed a measure of exposure to shifts in trade policy and provided its economic motivation. We now want to test whether our ex-ante measure captures cross-sectional variation in reactions to trade policy shifts observed ex-post. We expect that firms with a high score will suffer relatively more when trade barriers increase, while firms with a low

score will tend to benefit relatively more from an increase in trade barriers. To facilitate our discussion, we refer to firms with a high score as ‘trade peace’ firms, and firms with a low score as ‘trade war’ firms.

3.1 | Stock price reactions as ex-post exposures

To test empirically whether our multidimensional score captures exposure to shifts in trade policy, we conduct an event study of reactions to trade policy events. Price reactions can be interpreted as an ex-post market-based measure of exposure.¹⁸ Following an event that brings relevant economic news, market participants will update their expectations, and price reactions reveal changes in market expectations over future firm cashflows and/or risk.¹⁹ Focusing on the period following the selected events allows us to link changes in market expectations to the common narrative that characterises the events, which in our case is a shift in trade policy. This means that price reactions to policy shifts represent an ex-post measure that captures market participants’ assessment of exposure of firms to the policy shift. In practical terms, given that our hypothesis is that trade peace stocks stand to lose more than trade war stocks when governments raise barriers to international trade, we test whether the stock price reactions of trade peace stocks to these events are significantly more negative than those of trade war stocks.

First, we define event days on which there is a clear change in trade policy. To be useful for us, we need events which are important enough to influence stock prices. This is a first obvious condition to be able to estimate the impact of trade events on stock prices. We follow Amiti et al. (2020), who propose an objective method to define events. They use peaks in Google searches for the term *trade war*, which match with clearly identifiable changes in trade policy. To ensure that we use the most influential events, we focus on the tariff events that characterise the trade war between the United States and China, the world’s two biggest economies. Peaks in Google searches indicate that the events receive much attention. The assumption that stock prices should be influenced by them is therefore reasonable. Table 3 presents the seven resulting event dates, together with a short description of the trade policy change.

We also need to specify the event window in which we expect the event to impact stock returns. The event dates define the moment when the news of a trade policy change reaches the market. Only analysing returns on this day, however, may not allow enough time for the market to fully adjust prices to the event. The event window should also not be too long since we want to avoid capturing price reactions to subsequent nontrade-related news. Therefore, we define the main event window to include the event date and the subsequent 7 calendar days. In the Supporting Information S1: Appendix, we show that conclusions are not sensitive to using event windows comprised of the event date and the subsequent day or the event date and the subsequent 14 days instead. The length of these event windows is in a similar range as those used by Davies and Studnicka (2018) and Amiti et al. (2020).

¹⁸This is similar to Greenland et al. (2020), although we emphasise that, as opposed to abnormal returns, our exposure measure is available ex-ante and not purely empirical but based on fundamental economic mechanisms. See also Fisman et al. (2014), Davies and Studnicka (2018), Egger and Zhu (2020) and Bianconi et al. (2021) for the use of event studies related to international trade.

¹⁹This follows from prices forming as the sum of the stock’s discounted future expected cashflows.

TABLE 3 Trade events.

This table reports the events in the US tariff events and Chinese retaliation events taken from Amiti et al. (2020).

Date	Event description
01 March 2018	US announces steel and aluminium tariffs
22 March 2018	US orders identification of Chinese products for tariffs
02 April 2018	China to impose tariffs on 128 US exports
15 June 2018	China retaliates on USD50bn of US imports
17 September 2018	US announces tariffs on USD200bn goods from China
10 May 2019	US raises tariffs from 10% to 25% on USD200bn of Chinese imports
23 August 2019	China raises tariffs on soy and autos

Then, we want to estimate the stock-specific impact of the events on the share price during the event windows. To isolate the stock-specific component of the stock returns, it is necessary to control for the factors driving returns independent of the event. Therefore, we estimate a model for each stock's expected returns based on daily returns from 1 year before each event until 1 week before the event.²⁰ Our main results are based on the classic CAPM as the model for normal returns, but in the Supporting Information S1: Appendix, we show that conclusions are similar when using the Fama and French (2015) five-factor model.

$$R_t^i = \alpha^i + \beta^i \times MKT_t + \varepsilon_t^i.$$

R_t^i represents the returns of stock i at day t in excess of the risk-free rate and MKT_t is the return in the market factor.

For each day t in the event windows, we can then define the abnormal return AR_t^i for stock i as the difference between the realised return RR_t^i and the normal expected return NR_t^i . In the equations below, RF_t is the risk-free rate on day t and $\hat{\beta}^i$ is the estimated value of β^i above.

$$AR_t^i = RR_t^i - NR_t^i, \text{ with } NR_t^i = RF_t + \hat{\beta}^i \times MKT_t.$$

Finally, the cumulative abnormal return CAR^i for stock i , over a given event window, is the sum of its abnormal returns on each day in that window.²¹ Since the event windows are chosen as moments on which shifts in trade policy are expected to influence stock prices and market-wide returns are controlled for, the cumulative abnormal returns capture the stock-specific impact of trade policy shifts. Therefore, we interpret it as an ex-post measure of exposure to such shifts.

²⁰If the period from 1 year until 1 week before an event includes other event windows, we exclude these event windows from the data when estimating the model for normal returns.

²¹We follow standard practice in event study and take the sum of daily returns as opposed to the product of cumulative daily returns. This allows us to conduct standard hypothesis for arithmetic averages (i.e., a t -test). Given the short time span of events (a few days), the difference between the two ways of computing cumulative returns over the event window will be tiny.

TABLE 4 Conditional average cumulative abnormal returns on event days.

This table reports the conditional average cumulative abnormal returns and *t*-statistics (in brackets). Cumulative abnormal returns are based on the CAPM model to estimate normal returns and an event window from the event day until 1 week after the event. The event dates are 01/03/2018, 22/03/2018, 02/04/2018, 15/06/2018, 17/09/2018, 10/05/2019 and 23/08/2019. Stock selections are made based on the augmented tradability, augmented export share, augmented risk disclosures, or combined scores as of the most recent June before each event. Stocks with a score above or equal to the 70th percentile are trade peace stocks, stocks with a score below or equal to the 30th percentile are trade war stocks. The *t*-statistics are adjusted for event-induced variance and correlation across stocks, following Kolari and Pynnönen (2010). The universe consists of the 1,000 largest US firms traded on the New York Stock Exchange or Nasdaq.

	Tradability	Export share	Risk disclosures	Combined
Trade peace stocks	−0.4% (−1.3)	−0.4% (−1.5)	−0.6% (−2.4)	−0.7% (−2.8)
Trade war stocks	0.3% (1.1)	0.7% (2.1)	0.7% (2.2)	0.7% (2.1)

$$CAR^i = \sum_{t \in \text{event window}} AR_t^i.$$

Our goal is to test whether our combined exposure measure, which is available before the events, is related to stock price reactions to the trade policy announcements, as estimated by the cumulative abnormal returns. A strong relation would validate our combined exposure measure. We analyse this relationship not only for the combined exposure measure, but also for the augmented tradability, export share and risk disclosures scores. The corresponding results will justify empirically why combining the different dimensions adds value.

A simple first test of this relationship is to compare the conditional average cumulative abnormal returns between trade peace and trade war firms. In this analysis, we define the 30% of firms with the highest and lowest exposure scores as trade peace and trade war firms. This selection is made as of the most recent June before each event date and the cumulative abnormal returns are pooled across events to calculate the conditional averages. Table 4 presents the corresponding results, based on one of the augmented or combined exposure scores.

These results could be interpreted from the perspective of investors that want to manage trade policy risks via an equal-weighted trade peace or trade war portfolio. In this interpretation, the results roughly represent the excess returns on the portfolios during one of the event windows, after controlling for market exposures. The average cumulative abnormal returns of trade peace stocks formed based on the individual dimensions range from −0.4% to −0.6%, while these values for the trade war stocks range from 0.3% to 0.7%. Based on *t*-statistics that account for event-induced variance and correlation structures between stocks (Kolari & Pynnönen, 2010), results are mixed in terms of statistical significance. Using the combined scores to create the stock selections, trade peace stocks have an average cumulative abnormal return of −0.7% and for trade war stocks this is 0.7%. Both values are statistically significant. Therefore, the combined exposure score allows to create well-diversified portfolios that generate a return spread of 1.4% over the short event windows, which consist of only 6 trading days. This value is economically meaningful. It is more than 10 times the long-term average of

the weekly returns of the US excess market returns.²² The effect is also large in comparison to the pre-announcement impact of the FOMC meetings, another set of events for which the literature has documented an important impact on market's returns. For instance, Lucca and Moench (2014) estimate that the average abnormal return on S&P 500 observed in the hours before the FOMC announcements is 0.5%,²³ which is similar in magnitude to the effect that we document for both the trade peace and the trade war portfolio. These results also show that the combined score adds value over the individual dimensions, since it results in a larger spread between trade peace and trade war stocks.

Next to the conditional averages, we also present results based on regressions with the cumulative abnormal returns as dependent variable, and the augmented and/or combined exposure measures as independent variables. The observations from the seven event dates are pooled together in one regression,²⁴ after controlling for event-specific returns by de-meaning the cumulative abnormal returns per event. For each stock and each event, the augmented and combined exposure scores available as of the most recent June before the event date are linked to the cumulative abnormal return. These exposure scores are standardised to a z-score per event, to facilitate interpretation. Regression coefficients therefore indicate the expected impact on the cumulative abnormal return of a one standard deviation increase in the exposure score. Since stocks with a high exposure score are trade peace stocks and the events represent an increase in trade barriers, we expect the regression coefficients to be negative. Note that any differences in market exposures across stocks are controlled for via the construction of the abnormal returns. We run the pooled regressions with and without a set of control variables, which includes the natural logarithm of stocks' market capitalisation and dummy variables indicating membership in 10 broad sectors.²⁵ This allows us to test whether our exposure measures add value in explaining cumulative abnormal returns over the control variables. Standard errors are estimated with a bootstrap methodology that accounts for industry clustering²⁶ and the estimation uncertainty of the covariance components in the regressors. This bootstrap methodology is detailed in the Supporting Information S1: Appendix.

Table 5 shows the results of several specifications of these regressions. In Panel A we report the results of the regressions without controls. The first, second and third specification use either the augmented tradability, export share or risk disclosures scores as a single independent variable. All resulting coefficients are significantly negative, ranging from -0.3% to -0.5% . This shows that each of the individual dimensions contains relevant information to capture a firm's exposure to trade policy shifts. The fourth regression includes all three measures as independent variables. Importantly, these results suggest that each dimension contains relevant information beyond what is already captured by the other two. All three coefficients

²²We estimate the average weekly excess equity market return in the U.S. using the data from the Kenneth French Data library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html) over the period from the 1st of July 1963 to the 30th of June 2020.

²³Lucca and Moench (2014) shows that the effect of FOMC announcements accrues mostly in the hours preceding the announcement and it does not revert afterwards.

²⁴We do not attempt to rank the importance of the shock of each event relative to the others. Instead, we focus on the regressions in which we pool the seven events together, which allow us to test whether we can capture the heterogeneity of a trade shock on average. However, for completeness, we also report in the Supporting Information S1: Appendix results of regressions in which we consider the seven events separately.

²⁵The sector classification is based on TRBC.

²⁶To define the industry clusters we use the OECD industry classification that underlies the construction of the export share characteristic.

TABLE 5 Relation between exposure scores and cumulative abnormal returns.

This table reports the coefficients, *t*-statistics (in brackets) and adjusted R^2 (in percent) values of regressions of percent cumulative abnormal returns on augmented tradability, augmented export share, augmented risk disclosures and/or combined scores. Cumulative abnormal returns are based on the CAPM model to estimate normal returns and an event window from the event day until 1 week after the event. The event dates are 01/03/2018, 22/03/2018, 02/04/2018, 15/06/2018, 17/09/2018, 10/05/2019 and 23/08/2019. Values for the independent variables as of the most recent June before each event are used. All event dates are pooled together in the regression, with cumulative abnormal returns de-meanned per event and regressors transformed to a *z*-score per event. Panel A reports results of the pooled regressions without control variables. Panel B reports results of pooled regressions in which we add as control variables the log-market cap (also standardised) and the dummy variables of the TRBC sector of the stocks. We obtain the *t*-statistics as the ratio of the coefficients of the regressions over standard errors that account for industry clustering (based on the OECD industry classification) and for the fact that the regressors are estimated. We compute standard errors with a bootstrapping methodology detailed in the Supporting Information S1: Appendix. The universe consists of the 1000 largest US firms traded on the New York Stock Exchange or Nasdaq.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A: Without control variables</i>								
Tradability	−0.3 (−2.9)			−0.3 (−2.3)		0.0 (0.1)		
Export share		−0.5 (−2.9)		−0.3 (−1.4)			0.0 (0.1)	
Risk disclosures			−0.5 (−5.0)	−0.3 (−2.1)				−0.1 (−0.4)
Combined					−0.6 (−8.7)	−0.6 (−7.2)	−0.6 (−2.5)	−0.6 (−4.5)
Adj. R^2	0.6	1.3	1.7	2.2	2.2	2.2	2.2	2.2
<i>Panel B: With control variables</i>								
Tradability	−0.2 (−1.7)			−0.1 (−1.2)		0.1 (0.8)		
Export share		−0.5 (−2.9)		−0.4 (−1.8)			−0.2 (−0.7)	
Risk disclosures			−0.4 (−3.7)	−0.2 (−1.5)				0.0 (0.2)
Combined					−0.5 (−4.6)	−0.6 (−3.7)	−0.3 (−1.6)	−0.5 (−3.0)
Adj. R^2	1.8	2.4	2.2	2.5	2.5	2.5	2.5	2.4

are negative and similar in magnitude to the coefficients in the univariate regressions, with two out of three statistically significant. The adjusted R^2 of the regression also increases to 2.2% from 0.6% to 1.7%, indicating that the three measures combined capture more of the variation in cumulative abnormal returns compared with each one individually. Based on regressions

one to four, it clearly seems that combining the information in the three dimensions adds value. Next, we test whether the combined measure is constructed in a way that maintains this potential combined explanatory power. The results in the fifth column show that this is the case, with the same adjusted R^2 value as for regression four. The combined measure thus explains as much variation in cumulative abnormal returns as the combination of the three individual dimensions. The regression coefficient is again highly significantly negative at -0.6% . A one standard deviation increase in the combined exposure score corresponds to a cumulative abnormal return that is 60 bp lower. The last three specifications test whether one of the individual dimensions can still add value once we control for the combined exposure score. However, the results in columns six to eight show that this is not the case. None of the coefficients for the individual dimensions is significantly different from zero, while the size of the coefficient for the combined score and the adjusted R^2 are unaffected. Together with the fact that all coefficients in regression four are the same size, this indicates that averaging the individual dimensions with an equal weight in the combined exposure score results in a measure that captures all relevant information in the three individual dimensions.

In panel B of Table 5 we report the results of the regressions in which we add our set of control variables. Overall, adding the control variables does not affect our results. In particular, we highlight that the coefficients of the regressions in which we consider our three measures of exposure separately are close to the ones reported in panel A, and two out of three are still statistically significant. More importantly, the coefficient of our combined measure in specification 5 is still almost as negative as the one reported in panel A and preserves a large t-statistic. This suggests that our combined measure of exposure adds independent information to capture variation in price reactions to trade policy shocks relative to firm size and sector membership. In addition, we also highlight that the adjusted R^2 in specification 5 of panel A (2.2%) and B (2.5%) are very close, which indicates that the control variables have little residual explanatory power over our combined measure.²⁷

To sum up, tradability, export share and risk disclosures each contain useful information to explain differences in stock price reactions to trade policy announcements, and the combined measure successfully captures all this information. In addition, the information provided by our measures of exposure cannot be fully captured by firm characteristics that leverage on limited information, such as size and sector membership.

3.2 | Ex-post exposures extracted from earnings calls

We propose a firm-level measure to capture exposure to shifts in trade policy and have so far shown its validity via ex-post observed abnormal returns around trade policy changes. Even though we show the robustness of results to some specification choices when estimating cumulative abnormal returns in the Supporting Information S1: Appendix, we still depend on

²⁷Furthermore, in the Supporting Information S1: Appendix, we formally show that our measures of exposure are preferable relative to the control variables to capture variation in cumulative abnormal returns, using LASSO regressions. In the Supporting Information S1: Appendix, we also show graphically that there is substantial dispersion in terms of combined measure of exposure within the 10 sectors. This means that our combined measure can capture differences in exposures to international trade that go beyond what is explained by differences in broad sector definitions and helps to explain the finding that sector dummy variables are dominated by our combined exposure measure in the LASSO regressions.

parameters such as the selection of events. Furthermore, returns are noisy, especially during these relatively short time windows. Therefore, to increase our confidence that the measure captures a fundamental economic exposure, it is useful to test whether it can also explain variation in alternative ex-post exposure measures. We perform similar regressions as those presented in Table 5, but instead of using cumulative abnormal returns as the dependent variable, we now use sentiment or uncertainty regarding international trade extracted from a firm's earnings calls.²⁸ These alternative ex-post exposure measures are based on a completely different type of data than abnormal returns. Results similar to those reported in the previous section will therefore strongly support the conclusion that the combined score reliably captures exposures to shifts in trade policy.

To construct firm-level measures of sentiment or uncertainty regarding international trade, we first collect all available quarterly earnings call transcripts for the firms in our universe from Refinitiv. Earnings calls typically include two sections. The firm's management will first provide a presentation discussing the previous quarter's results and their expectations going forward. Then, there is a Q&A section with external analysts. This data source therefore contains both internal and external perspectives on the issues facing the firm. From the transcript text, we want to extract ex-post measures of exposure to shifts in trade policy. Following Hassan et al. (2019), we assume that firms heavily impacted by changes in trade policy will devote a lot of time in their earnings call to discuss this.

Our analysis on trade sentiment relies on the further assumption that trade policy discussions with a positive connotation indicate that a firm is positively impacted by current changes in trade policy, while a negative connotation indicates the opposite. Note that in the construction of the risk disclosures characteristic, we implicitly assume that all topics discussed in a firm's risk disclosures impact the firm negatively. It does not seem reasonable to make the same assumption in the case of earnings calls, however, due to the different nature of this data source. Consequently, trade sentiment captures how much attention firms spend on trade policy in their earnings calls and whether these discussions have a positive or negative connotation. Trade uncertainty, on the other hand, aims to capture discussions related to trade policy that are characterised by a heightened sense of uncertainty. During periods in which many trade policy changes occur, either towards globalisation or towards protectionism, we expect that firms with a high trade policy exposure experience an increase in uncertainty regarding their future business environment, as captured by high trade uncertainty.

Our analysis of sentiment and uncertainty from earnings call transcript data provides a true out of sample test of the exposure measure used to obtain ex-ante information on trade policy exposure, since we do not use information from earnings call transcripts in constructing the ex-ante measure. Besides withholding data for an out of sample assessment there are more important reasons for why we do not include the textual data from earnings call transcript in our ex-ante measure of exposure. As we will show below, analysis of earnings call transcripts shows large variation in the attention given to the trade policy topic over time, with very little attention in early parts of the sample. These variations in attention reflect that discussions between management and analysts tend to pick up topics of current interest and high media attention. The low attention to trade policy topics in earnings calls over earlier parts of the sample implies that it would have been close to impossible to obtain reliable information on

²⁸Sentiment has also been used to study the heterogenous exposure to the impact of the Covid-19 outbreak in Ramelli and Wagner (2020).

exposure to trade policy risk ex-ante. Text from risk disclosures at the firm level, which we use in ex-ante measurement, does not display such an attention effect since firms provide a comprehensive list of relevant risk exposures at each point in time. In ex-post validation, the effects of time variation in attention are actually useful because one would expect high exposure firms to display negative sentiment and increased uncertainty ex-post during periods when attention to trade policy rises due to increased trade tensions. We provide further analysis to support our argument that earnings call transcripts are not useful for ex-ante exposure measurement in the Supporting Information S1: Appendix.

A dictionary-based approach similar to the one we used to construct the risk disclosures characteristic is suitable to capture how much attention firms spend on trade policy in their earnings calls but does not capture the sentiment or uncertainty in these discussions. Therefore, we follow Hassan et al. (2019) and Caldara et al. (2020) in their definition of measures that also capture the sentiment or uncertainty surrounding a given topic.

$$\text{Trade sentiment} = \frac{\sum_w^{W_T} (1_{w \in \text{keywords}} \times \sum_{c=w-10}^{w+10} S(c))}{W_T},$$

$$\text{Trade uncertainty} = \frac{\sum_w^{W_T} (1_{w \in \text{keywords}} \times 1_{\text{uncertainty} \in w-10:w+10})}{W_T}.$$

W_T is the total number of words in an earnings call, $1_{w \in \text{keywords}}$ takes the value of 1 if the term w is in the trade dictionary and 0 otherwise, $S(c)$ takes the value of +1 when c is in the positive word list from Loughran and McDonald (2011), −1 when c is in their negative word list and 0 otherwise, and $1_{\text{uncertainty} \in w-10:w+10}$ takes the value of 1 when at least one of the uncertainty terms used by Caldara et al. (2020) is in the range of 10 words before to 10 words after term w and 0 otherwise. The trade dictionary is the same as the one used to construct the risk disclosures characteristic, based on the trade-specific keywords proposed by Baker et al. (2016) and Caldara et al. (2020).²⁹

This procedure gives us quarterly, firm-level measures of trade sentiment and uncertainty, which we interpret as ex-post proxies for firms' exposure to shifts in trade policy. Trade sentiment captures first moment effects, indicating whether a firm is positively or negatively impacted by current changes in trade policy. Trade uncertainty captures second moment effects, indicating whether a firm is strongly impacted in absolute terms. Details on the procedure to create the trade sentiment and uncertainty measures, including text preprocessing, validation checks and tests with alternative specifications can be found in the Supporting Information S1: Appendix.

Before we do the regression analysis to test the relationship between our augmented or combined exposure scores and trade sentiment or uncertainty, it is important to point out that a specific earnings call tends to discuss a limited number of topics. In other words, not many firms will discuss trade policy when there are no big changes to discuss. Of course, to do a useful analysis, we require that a reasonably large sample of firms actually discuss trade policy in their

²⁹Similar to how we proceeded when using corporate risk disclosures, we do not count mentions of the keyword tariff if they appear to be referring to regulated prices instead of import and export duties. In particular, we do not count mentions of the keyword tariff when for telecom and utility firms, or when the word appears next to a set of exclusion terms indicating a context of regulated prices, as published by Caldara et al. (2020).

earnings calls. Otherwise, the trade sentiment or uncertainty measures will automatically have a value of zero for most firms in the universe and will thus not be informative. Figure 1 highlights this point by showing the time series of a measure of trade policy attention, defined as the percentage of earnings call transcripts linked to firms in our universe that mention at least one term in our trade dictionary. In most quarters until 2016, only around 5% of firms seem to discuss trade policy in their earnings calls. There is a first sharp peak to around 20% in the first quarter of 2017, before a longer period of increased attention between the second quarter of 2018 and the first quarter of 2020. To ensure we have enough firms that discuss trade policy in our sample, the regression analysis will be based on earnings calls from the third quarter of 2018. This is when trade policy attention peaks at around 35%.

This period of increased attention coincides with the tariff announcements on which our event study is based, that is, the midst of the US-China trade war. Therefore, we characterise this period as a period of increasing trade barriers. Consequently, in the third quarter of 2018, trade peace firms are expected to have a relatively more negative trade sentiment and trade war firms are expected to have a relatively more positive trade sentiment. Furthermore, trade peace firms, which are more exposed to international trade, are expected to have a higher trade uncertainty. In other words, the regression coefficients of trade sentiment and trade uncertainty on the augmented and combined exposure scores are expected to be negative and positive, respectively.

Table 6 presents the results for regressions of trade sentiment or trade uncertainty in the third quarter of 2018 on different measures of firm-level exposure to trade policy risk. Note that the augmented and combined scores are estimated ex-ante, as of June 2018, while sentiment and uncertainty play out over the following quarter. Both dependent and independent variables are transformed to a z-score, so the coefficients can be interpreted as the expected number of standard deviations increase in trade sentiment or uncertainty for a one standard deviation increase in exposure score.

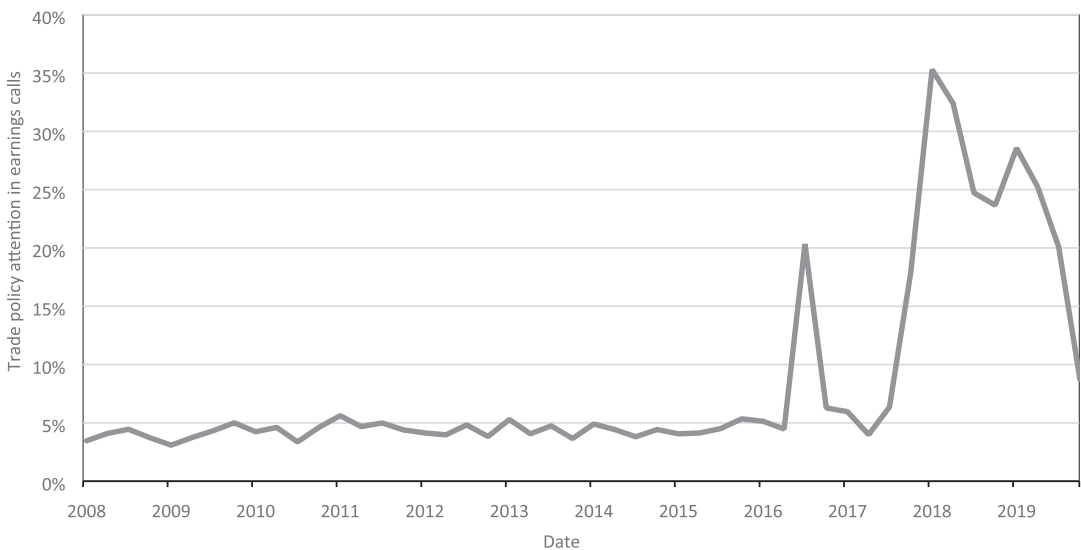


FIGURE 1 Trade policy attention in earnings calls. The graph shows the percentage of earnings calls that mention at least one term in our trade dictionary, in each quarter from Q3 2008 until Q2 2020. The universe consists of the 1000 largest US firms traded on the New York Stock Exchange or Nasdaq.

TABLE 6 Relation between exposure scores and trade sentiment or uncertainty.

Panel A of this table reports the coefficients, *t*-statistics (in brackets) and adjusted R^2 values of regressions of trade sentiment in earnings calls on augmented tradability, augmented export share, augmented risk disclosures and/or combined scores. Panel B of the table reports the coefficients, *t*-statistics (in brackets) and adjusted R^2 values of regressions of trade uncertainty in earnings calls on augmented tradability, augmented export share, augmented risk disclosures and/or combined scores. The analysis is based on earnings call transcripts from Q3 2018 and exposure scores available in June 2018. All variables are transformed to a *z*-score. *T*-statistics are based on a bootstrap methodology that accounts for industry clustering (based on the OECD industry classification) and estimation error in the regressors, detailed in the Supporting Information S1: Appendix. The universe consists of the 1000 largest US firms traded on the New York Stock Exchange or Nasdaq.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A: Trade sentiment</i>								
Tradability	0.03 (0.6)			0.06 (1.3)		0.13 (2.2)		
Export share		-0.11 (-2.8)		-0.04 (-0.8)			-0.08 (-1.0)	
Risk disclosures			-0.14 (-3.6)	-0.13 (-2.5)				-0.17 (-2.5)
Combined					-0.10 (-2.7)	-0.18 (-3.0)	-0.04 (-0.5)	0.04 (0.6)
Adj. R^2	0.0	1.2	1.9	2.2	1.0	2.1	1.2	1.8
<i>Panel B: Trade uncertainty</i>								
Tradability	-0.05 (-0.6)			-0.11 (-1.7)		-0.25 (-3.0)		
Export share		0.22 (3.3)		0.08 (1.0)			0.15 (1.3)	
Risk disclosures			0.28 (5.6)	0.26 (4.2)				0.34 (3.8)
Combined					0.21 (3.4)	0.35 (4.3)	0.10 (0.8)	-0.07 (-0.7)
Adj. R^2	0.2	4.9	7.8	9.3	4.4	8.7	5.2	7.8

The results in Panel A of Table 6 show a significant negative relation between the combined measure of trade policy exposure and trade-related sentiment. This finding implies that stocks identified as risky according to the combined measure face low sentiment in times of negative trade policy events, as would be expected. More precisely, the fifth column shows that the combined exposure score has a significantly negative coefficient. A one standard deviation increase in the combined score decreases the expected trade sentiment by around 10% of a standard deviation. However, the findings in Panel A also show that the expected relationship

is not confirmed for all the single dimensional exposure measures. The first three columns show that, while the augmented export share and risk disclosure scores have a significantly negative coefficient, the coefficient for augmented tradability is insignificant. The results in Panel B give a similar picture. The relation between the combined exposure score and trade uncertainty is significantly positive, as expected. The fifth column shows that a one standard deviation increase in ex-ante trade exposure corresponds to an increase in trade uncertainty by 21% of a standard deviation in this period of high trade tensions. Again, the first three columns indicate that the expected positive relationship holds for export share and risk disclosures, but ex-ante exposure based on tradability is not significantly related to trade uncertainty. These results suggest that using tradability alone, even though this is commonly done in the literature (e.g., Barrot et al., 2019), does not necessarily allow to capture trade-related sentiment or uncertainty. Combining the different single measures avoids this weakness of the tradability measure.

4 | INTERNATIONAL DATA

We next show that the combined exposure score is also useful to capture exposure to shifts in trade policy in an international stock universe. The results we have presented so far are all based on US data. Obtaining similar results with a completely different data set would again strengthen our confidence that we capture reliable economic mechanisms that will also be relevant going forward. In this section, we use data from Canada, countries in developed Europe, and countries in developed Asia Pacific. In total, we select a universe of around 1000 of the largest and most liquid firms in this region to form a good representation of the investable stock universe in the developed world outside the United States.³⁰

We define firms' tradability and export share characteristics in the same way as for US firms, as described above. Even though shipping costs are based on data from the US Census Bureau, it is reasonable to use the same values for other developed countries. The ranking of goods in terms of shipping cost is not expected to materially differ depending on the countries between which they are shipped. Indeed, Barrot et al. (2019) point out that shipping costs are tightly linked to the weight-to-value ratio of goods, which is clearly independent from the shipping route. Export share is based on data from multiple countries, as the definition above shows, so it is representative for our international universe. A big difference compared with US data, however, is that 10-K risk disclosures are only available for US firms. We have experimented whether it is possible to capture the desired ex-ante exposures from earnings call transcripts instead. As highlighted by Figure 1, trade policy is not often discussed when no big changes occur, so we concluded that this is not a feasible option.³¹ Therefore, our international exposure scores are only based on the tradability and export share dimensions. We augment the tradability and export share characteristics in mostly the same way as for the US firms. The only difference is that we use weekly instead of daily returns when estimating a stock's exposure to the mimicking portfolios, because of asynchronous returns in our international

³⁰We use data for the Scientific Beta Developed ex-US universe. The detailed construction process for this universe can be found in documentation accessible via www.scientificbeta.com.

³¹We present a more detailed explanation why we do not use earnings call transcripts for ex-ante exposure measurement in the Supporting Information S1: Appendix.

TABLE 7 Trade events for international data.

This table reports the 2018–2020 events based on the trade war timeline from Peterson Institute for International Economics.

Date	Event description
23 June 2016	UK votes to leave the EU in referendum
01 March 2018	US announces steel and aluminium tariffs
01 June 2018	US ends tariff exemptions for EU, Canada and Mexico
22 June 2018	EU retaliates with tariffs on US goods
01 July 2018	Canada imposes tariffs on US products
24 January 2020	US broadens steel and aluminium tariffs

universe. The combined exposure score is defined as the average of the augmented tradability and export share scores.

For the US data, we have shown that the combined measure captures exposure to shifts in trade policy, using abnormal returns around events that characterise the trade war between the United States and China as an ex-post proxy for these exposures. Of course, these events are less relevant for our international universe since neither US nor Chinese firms are included. To be able to do a similar analysis, we use the trade war timeline made available by the Peterson Institute for International Economics to define a set of relevant events.³² This timeline is based on a classification in five different battles. We focus on the one that is most relevant for the countries included in our international data, which is the series of actions around steel and aluminium. From all the events described there, we select all tariff announcements or the end of tariff exemptions that concern countries in our data. Furthermore, we decide to add the date of the Brexit referendum as a relevant event, following Davies and Studnicka (2018). The decision of the United Kingdom to leave the EU reduced confidence in the free trade agreements in place within the developed Europe region and increases the possibility of future trade barriers arising. Table 7 gives an overview of the selected events, which involve the countries and regions in our international universe.

For all stocks in the universe, we then obtain cumulative abnormal returns around these six events in mostly the same way as described for the US data. In this case, however, we estimate the CAPM model for normal returns using the past 2 years of weekly returns to account for asynchronous returns in our international universe. Abnormal returns on days around the event dates are then defined as the difference between realised and normal returns. Finally, the cumulative abnormal return is the sum of abnormal returns from the event date until 1 week after the event. Based on these cumulative abnormal returns as dependent variable, we run similar regressions as those shown in Table 5. The observations at each event date are again pooled together, after de-meaning the cumulative abnormal returns per event. The augmented tradability, augmented export share and combined scores available at the most recent June before each event date are used and standardised to a z-score per event.

³²See <https://www.piie.com/blogs/trade-investment-policy-watch/trump-trade-war-china-date-guide>. We use the version updated as of 13/03/2020.

TABLE 8 Relation between exposure scores and cumulative abnormal returns—International universe.

The table reports the coefficients, *t*-statistics (in brackets) and adjusted R^2 (in percent) values of regressions of percent cumulative abnormal returns on augmented tradability, augmented export share and/or combined scores. Cumulative abnormal returns are based on the CAPM model to estimate normal returns and an event window from the event day until 1 week after the event. The event dates are 23/06/2016, 01/03/2018, 01/06/2018, 22/06/2018, 01/07/2018 and 24/01/2020. Values for the independent variables as of the most recent June before each event are used. All event dates are pooled together in the regression, with cumulative abnormal returns de-meanned per event and regressors transformed to a *z*-score per event. We obtain the *t*-statistics as the ratio of the coefficients of the regressions over standard errors that account for industry clustering (based on the OECD industry classification) and for the fact that the regressors are estimated. We compute standard errors with a bootstrapping methodology detailed in the Supporting Information S1: Appendix. The universe consists of around 1000 of the largest and most liquid firms in a region consisting of Canada, developed Europe and developed Asia-Pacific.

	(1)	(2)	(3)	(4)	(5)	(6)
Tradability	−0.1 (−0.8)		−0.1 (−0.5)		0.3 (1.5)	
Export share		−0.4 (−2.8)	−0.4 (−2.7)			−0.3 (−1.5)
Combined				−0.4 (−2.2)	−0.6 (−2.8)	−0.1 (−0.5)
Adj. R^2	0.0	1.0	1.0	0.8	1.0	1.0

Table 8 shows the results of the analysis on international data. The first two columns show that tradability and export share are both negatively related to the cumulative abnormal returns, as expected, although only the coefficient for export share is statistically significant. As shown in column four, the combined score also has a significantly negative coefficient. A one standard deviation increase in the combined score corresponds to a cumulative abnormal return that is on average 40 bp lower. The significantly negative relationship between abnormal returns and the combined score supports the conclusion that our proposed measure is suitable to capture exposure to shifts in trade policy out-of-sample.

5 | CONCLUSION

Historically, there has been a long-term trend towards free international trade, but not without important reversals towards protectionism. Shifts in trade policy influence the structure of the economy, creating winners and losers. This paper proposes a multidimensional measure that captures firms' exposure to such shifts. We assess this measure of exposure via a double validation procedure using both stock returns and discussions between management and analysts. As a result, we document substantial heterogeneity in the impact of trade policy shifts across firms.

Our exposure measure is based on firm characteristics used in previous research. It combines the information captured by the tradability of the goods produced by a firm, the share of the output in the firm's industry that is exported, and the attention given to trade policy in the firm's official risk disclosures. Furthermore, we also make use of stock market information when constructing the

exposure measure. Each of the three characteristics is augmented with a covariance score that captures how similar a stock behaves compared with stocks with a high characteristic score. Adding covariances in the exposure measures allows us to extract information in stock returns that solely characteristics-based measures ignore. This multidimensional exposure measure also has the advantages that it is granular and available for all firms. Using a single characteristic to measure exposure would miss important information, since for many firms, characteristics are not available or are not able to discriminate between firms. Combining characteristics and extracting information from stock returns has the potential to lead to a more reliable measure of exposure.

We show that the exposure measure succeeds in capturing variation in exposure to shifts in trade policy out-of-sample via a double validation procedure that considers both price effects and information in earnings calls. To assess price effects, we use abnormal returns around tariff announcements in the US-China trade war as an ex-post proxy of firms' exposures. Our results show that trade war stocks have positive stock price reactions to these announcements while trade peace stocks react negatively. We further show that each dimension contains relevant incremental information and that the combined measure captures all this information. Next to being statistically significant, these results are also economically significant. Trade war and trade peace stocks exhibit average return differences of 140 bp over six trading days around the tariff announcements. Especially during periods of regular or large trade policy changes, these differences are important. In addition, we show that the multidimensional measure of exposure adds information beyond standard firm classifications and, beyond the US evidence, also captures return effects after trade policy shocks in international markets. We also analyse sentiment and uncertainty in discussions about trade policy during firms' quarterly earnings calls. Our multidimensional exposure measure is able to distinguish between firms with relatively positive and negative trade sentiment and between firms with high and low trade uncertainty during a period of turbulent trade policy. The fact that we obtain supportive evidence for our measure based on both stock returns and discussions during earnings calls strengthens our confidence that it reliably captures exposure to shifts in trade policy out-of-sample.

Overall, we find consistent evidence, both when analysing return effects and discussions between management and analysts, that the multidimensional measure reliably captures heterogeneity across firms around trade policy shocks. The measure adds value to other approaches in the literature because it is granular, is available for all firms, combines the information from multiple data sources, captures exposures out-of-sample, and is independent of specific trade policy changes. These attributes ensure that it is useful for a wide range of potential applications. Among others, it can be used in asset pricing to test whether exposure to trade policy risk is rewarded,³³ it allows to construct portfolios that hedge risk exposures in nontradable sources of investors' overall wealth, and it can add value to research that analyses reallocation effects of shifts in trade policy. We consider these applications to be interesting avenues for future research.

DATA AVAILABILITY STATEMENT

The main public data sources identified in the paper can be accessed via <https://www.census.gov/foreign-trade/data/index.html>, <https://www.oecd.org/industry/input-output-tables.htm> and <https://www.sec.gov/edgar/search-and-access>. Data licensing agreements prevent us from publicly sharing data on stock prices and earnings call transcripts.

³³We provide a preliminary analysis of this question in the Supporting Information S1: [Appendix](#).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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